

Recurrences II + Master (Decreasing)

More decreasing recurrences and the master theorem for decreasing functions.

Watch these Abdul Bari videos

- > 2.1.3 Recurrence Relation ($T(n) = T(n-1) + \log n$) #3
- > 2.1.4 Recurrence Relation $T(n) = 2T(n-1) + 1$ #4
- > 2.2 Masters Theorem Decreasing Function

Key concepts

- $T(n) = 2T(n-1) + 1 \rightarrow O(2^n)$.
- Master theorem (decreasing) gives quick answers.
- Identify a , b , $f(n)$ in the recurrence.

Key idea

$T(n) = 2T(n-1) + 1$ doubles each step $\rightarrow$ exponential.

Tasks (1.5h)

- ☐ Watch the 3 videos
- ☐ Solve the exponential recurrence
- ☐ Apply the decreasing master theorem

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0